

2. RELATIONS

Let S be the set of undergraduate students at some university and let C be the set of courses offered. Then the set $S \times C$ is the set of ordered pairs (s, c) where $s \in S$ and $c \in C$. Say we are only interested for the subset of $S \times C$ that only contains the set ordered pairs (s, c) with the property that the student s is registered for the course c . This subset captures the relation “The student s is registered for the course c ”.

We introduce now the concept of a binary relation to provide a mathematical description of any relationship between the elements of two sets.

Definition:

A **binary relation** R between two sets A and B is defined to be a subset of the Cartesian product $A \times B$. In the case where $A = B$, we simply refer to R as a **relation on A** .

Example 1:

Suppose that Fred and Helen are married with two children, Alice and Ken. Suppose that John and Mary are also married with three children, Sue, Mike and Penny. Suppose that Ken and Sue got married and had three children, Jane, Fiona and Alan.

Write down the ordered pairs belonging to the following relations on the set P of the people above.

(a) $R = \{(x, y): x \text{ is a grandfather of } y\}$

(b) $S = \{(x, y): x \text{ is a sister of } y\}$

Solution:

(a) $R = \{(\text{Fred}, \text{Jane}), (\text{Fred}, \text{Fiona}), (\text{Fred}, \text{Alan}), (\text{John}, \text{Jane}), (\text{John}, \text{Fiona}), (\text{John}, \text{Alan})\}$

(b) $S = \{(\text{Alice}, \text{Ken}), (\text{Sue}, \text{Mike}), (\text{Sue}, \text{Penny}), (\text{Penny}, \text{Sue}), (\text{Penny}, \text{Mike}), (\text{Jane}, \text{Fiona}), (\text{Jane}, \text{Alan}), (\text{Fiona}, \text{Jane}), (\text{Fiona}, \text{Alan})\}$

Both of these relations were relations on the same set.

Example 2:

Write down the ordered pairs belonging to the following binary relations between the sets $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6\}$.

(a) $U = \{(x, y): x + y = 9\}$

(b) $V = \{(x, y): x < y\}$

Solution:

(a) $U = \{(3, 6), (5, 4), (7, 2)\}$

(b) $V = \{(1, 2), (1, 4), (1, 6), (3, 4), (3, 6), (5, 6)\}$

Example 3:

Write down the ordered pairs belonging to the relation

$$R = \{(x, y): x \text{ is a divisor of } y\}$$

on the set $A = \{1, 2, 3, 4, 5, 6\}$.

Solution:

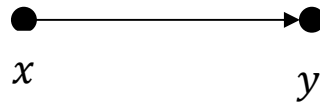
$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), \\ (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

We want to examine further ways to represent a given relation.

Digraphs:

Let R be a binary relation between two sets A and B . We represent all the elements of these two sets as the vertices of a graph. For each of the ordered pairs in the relation R , we draw an arrow linking the related elements. This is called a **directed graph** or **digraph**.

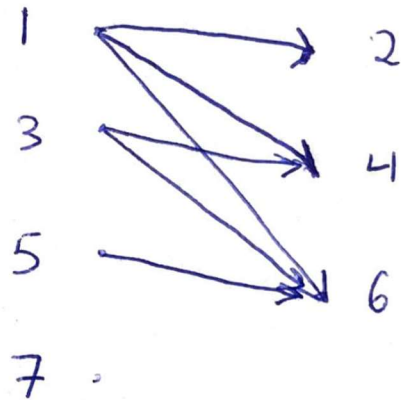
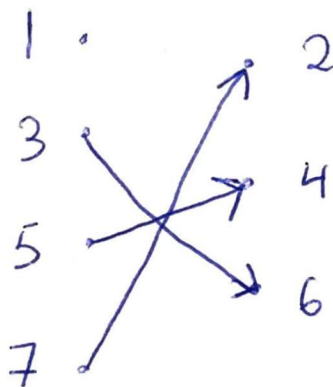
If for example (x, y) is an ordered pair in the relation R , then it is represented in the form:



Example 4:

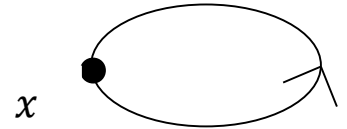
Draw the digraphs of the relations U and V from Example 2.

Solution:



For a relation on a single set A , we use a directed graph in which a single set of vertices represents the elements of A and arrows link the related elements.

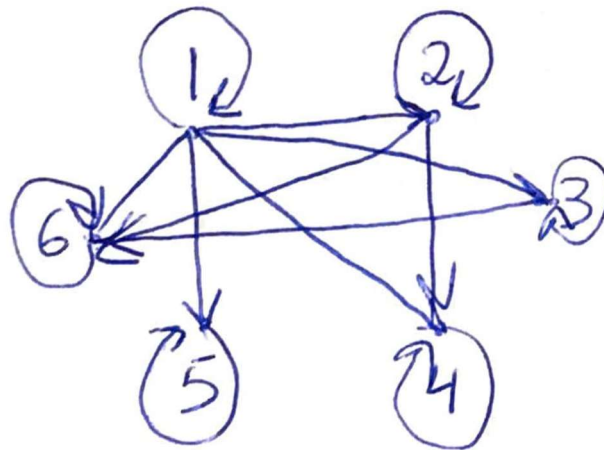
If (x, x) is an ordered pair in a relation on a set A , it is represented with a loop of the form:



Example 5:

Draw the diagram of the relation R from Example 3.

Solution:



Matrices

We can represent a relation R between two sets A (with cardinality $|A| = n$) and B (with cardinality $|B| = m$), with an array M with n rows and m columns. Such an array is called an **$n \times m$ matrix** (which reads “ **n by m matrix**”). The n rows are labelled by the elements of A in the order given and the m columns are labelled by the elements of B in the order given. The entry in row i and column j of the matrix is given by $M(i, j)$ where:

$$M(i, j) = T \quad \text{if} \quad (a_i, b_j) \in R$$

$$M(i, j) = F \quad \text{if} \quad (a_i, b_j) \notin R$$

Example 6:

Write down the matrix representations of the relations U and V from Example 2.

Solution:

$$U = \begin{bmatrix} F & F & F \\ F & F & T \\ F & T & F \\ T & F & F \end{bmatrix} \qquad V = \begin{bmatrix} T & T & T \\ F & T & T \\ F & F & T \\ F & F & F \end{bmatrix}$$

Example 7:

Write down the matrix representations of the relation R from Example 3.

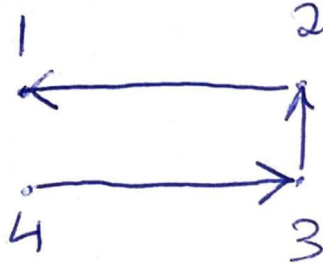
Solution:

$$R = \begin{bmatrix} T & T & T & T & T & T \\ F & T & F & T & F & T \\ F & F & T & F & F & T \\ F & F & F & T & F & F \\ F & F & F & F & T & F \\ F & F & F & F & F & T \end{bmatrix}$$

- If R is a binary relation between two sets we can write xRy whenever $(x, y) \in R$.
- To summarize, a binary relation between two finite sets may be described in any of the following ways:
 - (a) in words (using a predicate form)
 - (b) as a set of ordered pairs
 - (c) as a digraph (in graphical form)
 - (d) as a matrix (in matrix form)

Example 8:

The relation R on the set $A = \{1, 2, 3, 4\}$ has the following digraph representation.



List the ordered pairs belonging to R , write down its matrix representation and describe the relation in predicate form.

Solution:

$$R = \{(2, 1), (3, 2), (4, 3)\}$$

$$R = \begin{bmatrix} F & F & F & F \\ T & F & F & F \\ F & T & F & F \\ F & F & T & F \end{bmatrix}$$

$$R = \{(x, y) : x - y = 1\}$$

Example 9:

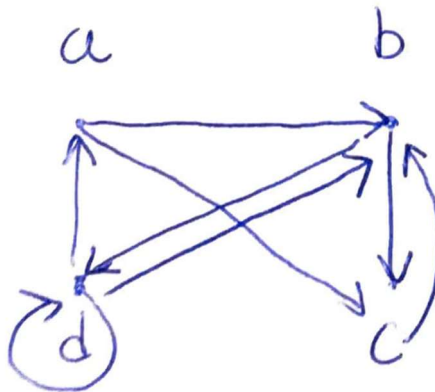
The relation R on the set $A = \{a, b, c, d\}$ has the following matrix representation.

$$R = \begin{bmatrix} F & T & T & F \\ F & F & T & T \\ F & T & F & F \\ T & T & F & T \end{bmatrix}$$

List the ordered pairs belonging to R and draw its digraph.

Solution:

$$R = \{(a, b), (a, c), (b, c), (b, d), (c, b), (d, a), (d, b), (d, d)\}$$



Properties of Relations

We restrict our attention to relations defined in a single set A and define the following properties:

A relation R on a set A is:

- **reflexive** when xRx for all $x \in A$.
In other words, a given relation R is reflexive if $(x, x) \in R$ for all possible values of x .
On a digraph, a relation is reflexive if there is always a loop from every vertex to itself.
- **symmetric** when if xRy then yRx for all x and y in A .
In other words, a given relation R is symmetric if when $(x, y) \in R$ then $(y, x) \in R$ for all x and y in A .
On a digraph, a relation is symmetric if whenever there is an edge from x to y there is also an edge from y to x .
- **antisymmetric** when if xRy and yRx then $x = y$ for all x and y in A .
In other words, a given relation R is antisymmetric if when $(x, y) \in R$ and $x \neq y$, then $(y, x) \notin R$.
On a digraph, a relation is antisymmetric if there is at most one edge between any pair x and y .
- **transitive** when xRy and yRz then xRz for all x, y and z in A .
In other words, a given relation R is transitive if when $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$.
On a digraph, a relation is transitive if whenever there is an edge from x to y and an edge from y to z , there is also an edge from x to z .

Example 10:

Consider the relation

$$R = \{(a, a), (a, b), (b, c), (a, c), (a, d), (b, d), (c, d), (b, b), (c, c)\}$$

on the set $A = \{a, b, c, d\}$. Determine if R is reflexive, symmetric, antisymmetric or transitive.

Solution:

$(d, d) \notin R \Rightarrow$ the relation is not reflexive.

$(a, b) \in R$ and $(b, a) \notin R \Rightarrow$ the relation is not symmetric.

The relation is antisymmetric.

$(a, b) \in R, (b, c) \in R$ and $(a, c) \in R$.

$(a, b) \in R, (b, d) \in R$ and $(a, d) \in R$.

$(b, c) \in R, (c, d) \in R$ and $(b, d) \in R$.

Therefore, the relation is transitive.

Definition:

A relation R on a set A that is reflexive, symmetric and transitive is called an **equivalence relation**.

Definition:

A relation R on a set A that is reflexive, antisymmetric and transitive is called a **partial order**.

Example 11:

Determine whether each of the relations is reflexive, symmetric, antisymmetric or transitive. Hence, determine if each of the relations is an equivalence relation, a partial order or neither.

- (a) $R = \{(x, y): x \text{ is a divisor of } y\}$ on the set of natural numbers.
- (b) $T = \{(x, y): x \neq y\}$ on the set of integers.
- (c) $U = \{(x, y): x \text{ has the same age as } y\}$ on the set of all people.
- (d) $V = \{(x, y): xy \geq 0\}$ on the set of integers.

Solution:

- (a) Every natural number is a divisor of itself; hence the relation is reflexive.

The relation is not symmetric. For example, 2 is divisor of 4, but 4 is not a divisor of 2. In other words, $(2, 4) \in R$ but $(4, 2) \notin R$.

The relation is antisymmetric since, x is a divisor of y and y is a divisor of x , if and only if $x = y$.

The relation is transitive. Suppose that x is a divisor of y , then we can write $y = mx$ (for some integer m) and that y is a divisor of z , then we can write $z = ny$ (for some integer n). Then $z = ny = n(mx) = (nm)x$. Hence, x is a divisor of z .

Therefore, the relation is a PARTIAL ORDER.

- (b) If $(x, x) \in T$ then $x \neq x$, which cannot be true for any $x \in \mathbb{Z}$.

Therefore the relation is not reflexive.

If $x \neq y$ then $y \neq x$, for all $x, y \in \mathbb{Z}$. Hence, if $(x, y) \in T$ then $(y, x) \in T$ as well. Hence, the relation is symmetric.

The relation is not antisymmetric for the same reason as above.

The relation is not transitive. For example, let $x = 1, y = 2$ and $z = 1$. Then $(1, 2) \in T, (2, 1) \in T$ but $(1, 1) \notin T$.

The relation is NEITHER an equivalence relation or a partial order.

(c) The relation is reflexive since any person x has the same age as themselves.

The relation is symmetric since, if x has the same age as y then y has the same age as x .

The relation is not antisymmetric since two different people can have the same age.

If a person x has the same age as a person y and y has the same age as z , then x has the same age as z . Hence, the relation is transitive.

The relation is an EQUIVALENCE RELATION.

(d) If $x = 0 \Rightarrow xx = 0$. If $x \neq 0 \Rightarrow xx = x^2 > 0$. In both cases $(x, x) \in V$. Hence, the relation is reflexive.

Since $xy = yx \Rightarrow$ if $xy \geq 0$ then $yx \geq 0$ for all integers x, y .

Hence, the relation is symmetric.

The relation is not antisymmetric. For example, let $x = 1$ and $y = 2$. Then $xy \geq 0$ and $yx \geq 0$ but $x \neq y$. In other words: $(1, 2) \in V$ and $(2, 1) \in V$ but $1 \neq 2$.

Let $x = -1, y = 0$ and $z = 1$. Then $xy = 0 = yz$ but $xz = -1 < 0$. In other words, $(-1, 0) \in V, (0, 1) \in V$ but $(-1, 1) \notin V$.

Hence, the relation is not transitive.

The relation is NEITHER an equivalence relation or a partial order.

Closure of a property

Definition:

If a relation R on a set A does not possess a property P , it may be possible to extend it to a relation R^* which does. We add to R additional ordered pairs so that the resulting relation R^* possesses the property P . If R^* is the smallest such set for which the property P holds, then R^* is called the **closure** of R with respect to the property P .

Example 12:

Let $A = \{1,2,3\}$ and R be the following relation on A :

$$R = \{(1,1), (1,2), (1,3), (3,1), (2,3)\}$$

The relation is not reflexive, not symmetric and not transitive. Find the corresponding closures.

Solution:

The closure with respect to the reflexive property must contain all pairs of the form (x, x) . Hence, the required closure is:

$$R^* = \{(1,1), (1,2), (1,3), (3,1), (2,3), \underline{(2,2)}, \underline{(3,3)}\}$$

The closure with respect to the symmetric property must contain the symmetrical counterpart of every pair in R . Hence, the required closure is:

$$R^* = \{(1,1), (1,2), (1,3), (3,1), (2,3), \underline{(2,1)}, \underline{(3,2)}\}$$

Note: if for example the pair $(1,2)$ was not in R , we wouldn't need to add the pair $(2,1)$.

Finding the transitive closure is a little trickier and might require a series of steps:

Since $(3,1) \in R$ and $(1,2) \in R$ then $(3,2)$ should be an element of R^* .

Since $(2,3) \in R$ and $(3,1) \in R$ then $(2,1)$ should be an element of R^* .

Since $(3,1) \in R$ and $(1,3) \in R$ then $(3,3)$ should be an element of R^* .

Adding these to R gives the relation:

$$R^* = \{(1,1), (1,2), (1,3), (3,1), (2,3), \underline{(3,2)}, \underline{(2,1)}, \underline{(3,3)}\}$$

But now, R contains $(2,1)$ and $(1,2)$, hence $(2,2)$ should be an element of R^* .

Hence, the transitive closure of R is:

$$R^* = \{(1,1), (1,2), (1,3), (3,1), (2,3), \underline{(3,2)}, \underline{(2,1)}, \underline{(3,3)}, \underline{(2,2)}\}$$

Note: If a relation is not antisymmetric, it means that it contains ordered pairs of the form (x, y) and (y, x) where $x \neq y$. Adding ordered pairs to R will not change this. Hence, there is no such thing as the antisymmetric closure.

Hasse Diagram

Remember that a partial order is a relation that is reflexive, antisymmetric and transitive. Partial orders are important in situations that we want to characterize precedence (decide whether one element of the set precedes another). Sets on which partial orders are defined are called **posets**.

If R is a partial order on a set A and $xRy, x \neq y$, we call x a **predecessor** of y , or equivalently, y is a **successor** of x . A given element y may have many predecessors. If, however, x is a predecessor of y and there is no other element z for which xRz and zRy , we call x an **immediate predecessor** of y and write $x < y$.

Elements with no predecessors are called **minimal elements**.

Elements with no successors are called **maximal elements**.

We can represent immediate predecessors on a graph known as the **Hasse diagram**. The vertices of the Hasse diagram represent the elements of the poset A , and if $x < y$, the vertex x is placed below the vertex y and the two vertices are joined by an edge. Hence, a Hasse diagram is read from bottom to top.

The Hasse diagram contains complete information about the original partial order provided we infer which elements are predecessors of others by working upwards through any chain of edges.

Note, that in contrast with the digraph:

- loops are not shown, since they are guaranteed to exist by the reflexive property
- edges guaranteed by the transitive property are not shown.

Example 13:

Let $R = \{(x, y): x \text{ is a divisor of } y\}$ be a partial order on the set $A = \{1, 2, 3, 6, 12, 18\}$. Draw a table of predecessors and immediate predecessors. Hence, draw the corresponding Hasse diagram.

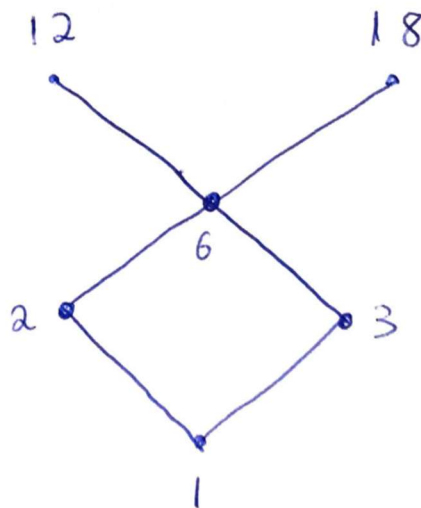
Solution:

Element	Predecessors	Immediate Predecessors
1	none	none
2	1	1
3	1	1
6	1, 2, 3	2, 3
12	1, 2, 3, 6	6
18	1, 2, 3, 6	6

Minimal Elements: 1

Maximal Elements: 12, 18

Hasse diagram:



Inverse Relation

Definition:

Given a binary relation R between two sets A and B , define the **inverse relation** R^{-1} between B and A by:

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

For example, if R is the relation “*is a parent of*” on the set of all people, then the inverse relation R^{-1} is the relation “*is a child of*”.

Example 14:

Let A and B be the sets $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6\}$. Let R be the relation

$$R = \{(x, y) : x < y\}$$

Describe the inverse relation R^{-1} between the sets B and A and enumerate its elements.

Solution:

First, we enumerate the elements of R :

$$R = \{(1, 2), (1, 4), (1, 6), (3, 4), (3, 6), (5, 6)\}$$

Then: $R^{-1} = \{(x, y) : x > y\}$ and

$$R^{-1} = \{(2, 1), (4, 1), (6, 1), (4, 3), (6, 3), (6, 5)\}$$

Notes:

- In graphical terms the inverse of a binary relation is obtained by reversing all the arrows in the digraph representation of the original relation.
- If R is a partial order, then R^{-1} is also a partial order (There is a formal proof of this statement, showing that if R is reflexive, antisymmetric and transitive, then so is R^{-1}). The minimal elements of R are the maximal elements of R^{-1} and the maximal elements of R are the minimal elements of R^{-1} .

Composition of two Relations

Definition:

Let R be a binary relation between two sets A and B , and S be a binary relation between B and a third set C . The **composition** of R and S , denoted by $S \circ R$, is the binary relation between A and C given by:

$$S \circ R = \{(a, c): a \in A, c \in C \text{ and } (aRb \text{ and } bRc) \text{ for some } b \in B\}$$

The new relation relates elements of A and C using elements of B as intermediates.

Example 15:

Let R be the relation “*is a sister of*” and S be the relation “*is a mother of*” on the set of all people. Describe the relations $S \circ R$ and $S \circ S$.

Solution:

$S \circ R$ is the relation “*is an aunt of*”

$S \circ S$ is the relation “*is a grandmother of*”

Example 16:

Let $A = \{a, b\}$, $B = \{1, 2, 3\}$ and $C = \{x, y\}$. Let R and S be relations from A and B and from B to C respectively as follows:

$$R = \{(a, 1), (a, 2), (a, 3), (b, 2)\}, \quad S = \{(1, y), (2, x), (3, x)\}$$

Enumerate the elements of R^{-1} , S^{-1} , $S \circ R$, $R^{-1} \circ S^{-1}$ and $(S \circ R)^{-1}$ and draw the corresponding digraphs.

Solution:

$$R^{-1} = \{(1, a), (2, a), (3, a), (2, b)\} \quad S^{-1} = \{(y, 1), (x, 2), (x, 3)\}$$

$$S \circ R = \{(a, y), (a, x), (b, x)\} \quad (S \circ R)^{-1} = \{(y, a), (x, a), (x, b)\}$$

$$R^{-1} \circ S^{-1} = \{(y, a), (x, a), (x, b)\}$$

Note:

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$$